

```
In [1]: import warnings
warnings.filterwarnings("ignore", category=UserWarning)
import numpy as np
from tqdm import tqdm
import torch
import torch.nn as nn
import torchani
import matplotlib.pyplot as plt
import sys
```

Use GPU

```
In [2]: device = torch.device('cuda' if torch.cuda.is_available() else 'cpu')
print(device)
print(torch.cuda.is_available())
print(torch.cuda.device_count())
print(sys.executable)
!queue -j $SLURM_JOB_ID -o "%.18i %.12P %.20b %.10T %.20N"
!nvidia-smi -L
```

cuda

True

1

/global/scratch/users/ericwangyz/chem242/ani/bin/python

	JOBID	PARTITION	TRES_PER_NODE	STATE	NODELIST
	33812409	savio3_gpu	gres:gpu:GTX2080TI:1	RUNNING	n0159.savio3

GPU 0: NVIDIA GeForce RTX 2080 Ti (UUID: GPU-33817832-42d8-7bb4-be03-bb4b50c253f1)

Set up AEV computer

AEV: Atomic Environment Vector (atomic features)

Ref: Chem. Sci., 2017, 8, 3192

```
In [3]: def init_aev_computer():
    Rcr = 5.2
    Rca = 3.5
    EtaR = torch.tensor([16], dtype=torch.float, device=device)
    ShfR = torch.tensor([
        0.900000, 1.168750, 1.437500, 1.706250,
        1.975000, 2.243750, 2.512500, 2.781250,
        3.050000, 3.318750, 3.587500, 3.856250,
```

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        4.125000, 4.393750, 4.662500, 4.931250
    ], dtype=torch.float, device=device)

    EtaA = torch.tensor([8], dtype=torch.float, device=device)
    Zeta = torch.tensor([32], dtype=torch.float, device=device)
    ShfA = torch.tensor([0.90, 1.55, 2.20, 2.85], dtype=torch.float, device=device)
    ShfZ = torch.tensor([
        0.19634954, 0.58904862, 0.9817477, 1.37444680,
        1.76714590, 2.15984490, 2.5525440, 2.94524300
    ], dtype=torch.float, device=device)

    num_species = 4
    aev_computer = torchani.AEVComputer(
        Rcr, Rca, EtaR, ShfR, EtaA, Zeta, ShfA, ShfZ, num_species
    )
    return aev_computer

aev_computer = init_aev_computer()
aev_dim = aev_computer.aev_length
print(aev_dim)

```

384

Prepare dataset & split

```

In [4]: def load_anl_dataset(dspath):
    self_energies = torch.tensor([
        0.500607632585, -37.8302333826,
        -54.5680045287, -75.0362229210
    ], dtype=torch.float, device=device)
    energy_shifter = torchani.utils.EnergyShifter(None)
    species_order = ['H', 'C', 'N', 'O']

    dataset = torchani.data.load(dspath)
    dataset = dataset.subtract_self_energies(energy_shifter, species_order)
    dataset = dataset.species_to_indices(species_order)
    dataset = dataset.shuffle()
    return dataset

dataset = load_anl_dataset("./ani_gdb_s01_to_s04.h5")
# Use dataset.split method to do split
train_data, val_data, test_data = dataset.split(0.8, 0.1, 0.1)

```

Batching

```
In [5]: batch_size = 8192
# use dataset.collate(...).cache() method to do batching
train_data_loader = train_data.collate(batch_size).cache()
val_data_loader = val_data.collate(batch_size).cache()
test_data_loader = test_data.collate(batch_size).cache()
```

Torchani API

```
In [6]: class AtomicNet(nn.Module):
    def __init__(self):
        super().__init__()
        self.layers = nn.Sequential(
            nn.Linear(384, 128),
            nn.ReLU(),
            nn.Linear(128, 1)
        )

    def forward(self, x):
        return self.layers(x)

class AtomicNetNarrowing(nn.Module):
    def __init__(self):
        super().__init__()
        self.layers = nn.Sequential(
            nn.Linear(384, 128),
            nn.ReLU(),
            nn.Linear(128, 64),
            nn.ReLU(),
            nn.Linear(64, 1)
        )

    def forward(self, x):
        return self.layers(x)

class AtomicNetBigger(nn.Module):
    def __init__(self):
        super().__init__()
        self.layers = nn.Sequential(
```

```

        nn.Linear(384, 256),
        nn.ReLU(),
        nn.Linear(256, 128),
        nn.ReLU(),
        nn.Linear(128, 1)
    )

    def forward(self, x):
        return self.layers(x)

class AtomicNetPaper(nn.Module):
    def __init__(self):
        super().__init__()
        self.layers = nn.Sequential(
            nn.Linear(384, 128),
            nn.ReLU(),
            nn.Linear(128, 128),
            nn.ReLU(),
            nn.Linear(128, 64),
            nn.ReLU(),
            nn.Linear(64, 1)
        )

    def forward(self, x):
        return self.layers(x)

class AtomicNetPaperWide(nn.Module):
    def __init__(self):
        super().__init__()
        self.layers = nn.Sequential(
            nn.Linear(384, 256),
            nn.ReLU(),
            nn.Linear(256, 256),
            nn.ReLU(),
            nn.Linear(256, 128),
            nn.ReLU(),
            nn.Linear(128, 1)
        )

    def forward(self, x):
        return self.layers(x)

class AtomicNetDropout(nn.Module):
    def __init__(self):

```

```

    super().__init__()
    self.layers = nn.Sequential(
        nn.Linear(384, 128),
        nn.ReLU(),
        nn.Dropout(p=0.1),
        nn.Linear(128, 64),
        nn.ReLU(),
        nn.Dropout(p=0.1),
        nn.Linear(64, 1)
    )

    def forward(self, x):
        return self.layers(x)

net_H = AtomicNet()
net_C = AtomicNet()
net_N = AtomicNet()
net_O = AtomicNet()

# ANI model requires a network for each atom type
# use torch.ANIModel() to compile atomic networks
ani_net = torchani.ANIModel([net_H, net_C, net_N, net_O])
model = nn.Sequential(
    aev_computer,
    ani_net
).to(device)

```

```

In [7]: train_data_batch = next(iter(train_data_loader))

loss_func = nn.MSELoss()
species = train_data_batch['species'].to(device)
coords = train_data_batch['coordinates'].to(device)
true_energies = train_data_batch['energies'].to(device).float()
_, pred_energies = model((species, coords))
loss = loss_func(true_energies, pred_energies)
print(loss)

```

tensor(0.3848, device='cuda:0', grad_fn=<MseLossBackward0>)

```

In [8]: class ANITrainer:
    def __init__(self, model, batch_size, learning_rate, epoch, l2):
        self.model = model

        num_params = sum(item.numel() for item in model.parameters())
        print(f"{model.__class__.__name__} - Number of parameters: {num_params}")

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self.batch_size = batch_size
self.optimizer = torch.optim.Adam(model.parameters(), learning_rate, weight_decay=l2)
self.epoch = epoch

def train(self, train_data, val_data, early_stop=True, draw_curve=True):
    self.model.train()

    # init data loader
    print("Initialize training data...")
    train_data_loader = train_data.collate(self.batch_size).cache()

    # definition of loss function: MSE is a good choice!
    loss_func = nn.MSELoss()

    # record epoch losses
    train_loss_list = []
    val_loss_list = []
    lowest_val_loss = np.inf

    for i in tqdm(range(self.epoch), leave=True):
        train_epoch_loss = 0.0
        for train_data_batch in train_data_loader:

            # compute energies
            species = train_data_batch['species'].to(device)
            coords = train_data_batch['coordinates'].to(device)
            true_energies = train_data_batch['energies'].to(device).float()
            _, pred_energies = self.model((species, coords))

            # compute loss
            batch_loss = loss_func(true_energies, pred_energies)

            # do a step
            self.optimizer.zero_grad() #clears previous batch gradient
            batch_loss.backward() #backprop
            self.optimizer.step()

            batch_importance = len(species) / len(train_data)
            train_epoch_loss += batch_loss.item() * batch_importance

        # use the self.evaluate to get loss on the validation set
        val_epoch_loss = self.evaluate(val_data)

        # append the losses

```

```

train_loss_list.append(train_epoch_loss)
val_loss_list.append(val_epoch_loss)

    if early_stop:
        if val_epoch_loss < lowest_val_loss:
            lowest_val_loss = val_epoch_loss
            weights = self.model.state_dict()

if draw_curve:
    fig, ax = plt.subplots(1, 1, figsize=(5, 4), constrained_layout=True)
    ax.set_yscale("log")
    # Plot train loss and validation loss
    ax.plot(np.arange(self.epoch), train_loss_list, label='Train')
    ax.plot(np.arange(self.epoch), val_loss_list, label='Validation')
    ax.legend()
    ax.set_xlabel("# Batch")
    ax.set_ylabel("Loss")

if early_stop:
    self.model.load_state_dict(weights)

return train_loss_list, val_loss_list

def evaluate(self, data, draw_plot=False):

    # init data loader
    data_loader = data.collate(self.batch_size).cache()

    # init loss function
    loss_func = nn.MSELoss()
    total_loss = 0.0

    if draw_plot:
        true_energies_all = []
        pred_energies_all = []

    with torch.no_grad():
        for batch_data in data_loader:

            # compute energies
            species = batch_data['species'].to(device)
            coords = batch_data['coordinates'].to(device)
            true_energies = batch_data['energies'].to(device).float()
            _, pred_energies = self.model((species, coords))

```

```

    # compute loss
    batch_loss = loss_func(true_energies, pred_energies)

    batch_importance = len(species) / len(data)
    total_loss += batch_loss.item() * batch_importance

    if draw_plot:
        true_energies_all.append(true_energies.detach().cpu().numpy().flatten())
        pred_energies_all.append(pred_energies.detach().cpu().numpy().flatten())

if draw_plot:
    true_energies_all = np.concatenate(true_energies_all)
    pred_energies_all = np.concatenate(pred_energies_all)
    # Report the mean absolute error
    # The unit of energies in the dataset is hartree
    # please convert it to kcal/mol when reporting the mean absolute error
    # 1 hartree = 627.5094738898777 kcal/mol
    # MAE = mean(|true - pred|)
    hartree2kcalmol = 627.5094738898777
    mae = np.mean(np.abs(true_energies_all - pred_energies_all)) * hartree2kcalmol
    fig, ax = plt.subplots(1, 1, figsize=(5, 4), constrained_layout=True)
    ax.scatter(true_energies_all, pred_energies_all, label=f"MAE: {mae:.2f} kcal/mol", s=2)
    ax.set_xlabel("Ground Truth")
    ax.set_ylabel("Predicted")
    xmin, xmax = ax.get_xlim()
    ymin, ymax = ax.get_ylim()
    vmin, vmax = min(xmin, ymin), max(xmax, ymax)
    ax.set_xlim(vmin, vmax)
    ax.set_ylim(vmin, vmax)
    ax.plot([vmin, vmax], [vmin, vmax], color='red')
    ax.legend()

return total_loss

```

In [9]: `break` #used when initializing everything in the notebook

Cell In[9], line 1

```

    bla bla break
    ^

```

SyntaxError: invalid syntax. Perhaps you forgot a comma?

Test on small data set for confirmation that everything working as expected


```
In [ ]: small_train = train_data.split(0.05)[0]
small_val = val_data.split(0.1)[0]
trainer = ANITrainer(model, batch_size=256, learning_rate=1e-3, epoch=10, l2=1e-5)
train_losses, val_losses = trainer.train(small_train, small_val)
```

Architecture tuning

```
In [ ]: #20% data on basic model, first will compare differing layer architectures
exp_train = train_data.split(0.20)[0]
exp_val = val_data.split(0.1)[0]

ani_net = torchani.ANIModel([AtomicNet(), AtomicNet(), AtomicNet(), AtomicNet()])
model = nn.Sequential(aev_computer, ani_net).to(device)

trainer = ANITrainer(model, batch_size=512, learning_rate=1e-3, epoch=30, l2=1e-5)
train_losses, val_losses = trainer.train(exp_train, exp_val)
test_loss = trainer.evaluate(test_data, draw_plot=True)
print(f"20% data test loss: {test_loss:.6f}")
#trial 1: 20% data test loss: 0.000019
#trial 2: 20% data test loss: 0.000020
#trial 3: 20% data test loss: 0.000015
#trial 4: 20% data test loss: 0.000014
#trial 5: 20% data test loss: 0.000019
```

```
In [ ]: exp_train = train_data.split(0.20)[0]
exp_val = val_data.split(0.1)[0]

ani_net = torchani.ANIModel([AtomicNetNarrowing(), AtomicNetNarrowing(), AtomicNetNarrowing(), AtomicNetNarrowing()])
model = nn.Sequential(aev_computer, ani_net).to(device)

trainer = ANITrainer(model, batch_size=512, learning_rate=1e-3, epoch=30, l2=1e-5)
train_losses, val_losses = trainer.train(exp_train, exp_val)
test_loss = trainer.evaluate(test_data, draw_plot=True)
print(f"Narrowing network test loss: {test_loss:.6f}")
#trial 1: Narrowing network test loss: 0.000023
#trial 2: Narrowing Network test loss: 0.000018
#trial 3: Narrowing network test loss: 0.000016
#trial 4: Narrowing network test loss: 0.000016
#trial 5: Narrowing network test loss: 0.000017
```

```
In [ ]: exp_train = train_data.split(0.20)[0]
exp_val = val_data.split(0.1)[0]
```

```

ani_net = torchani.ANIModel([AtomicNetBigger(), AtomicNetBigger(), AtomicNetBigger(), AtomicNetBigger()])
model = nn.Sequential(aev_computer, ani_net).to(device)

trainer = ANITrainer(model, batch_size=512, learning_rate=1e-3, epoch=30, l2=1e-5)
train_losses, val_losses = trainer.train(exp_train, exp_val)
test_loss = trainer.evaluate(test_data, draw_plot=True)
print(f"Bigger Network test loss: {test_loss:.6f}")
#trial 1: Bigger Network test loss: 0.000020
#trial 2: Bigger Network test loss: 0.000018
#trial 3: Bigger Network test loss: 0.000019
#trial 4: Bigger Network test loss: 0.000015
#trial 5: Bigger Network test loss: 0.000017

```

```

In [ ]: exp_train = train_data.split(0.20)[0]
exp_val = val_data.split(0.1)[0]

ani_net = torchani.ANIModel([AtomicNetPaper(), AtomicNetPaper(), AtomicNetPaper(), AtomicNetPaper()])
model = nn.Sequential(aev_computer, ani_net).to(device)

trainer = ANITrainer(model, batch_size=512, learning_rate=1e-3, epoch=30, l2=1e-5)
train_losses, val_losses = trainer.train(exp_train, exp_val)
test_loss = trainer.evaluate(test_data, draw_plot=True)
print(f"Paper Network test loss: {test_loss:.6f}")
#trial 1: Paper Network test loss: 0.000018
#trial 2: Paper Network test loss: 0.000018
#trial 3: Paper Network test loss: 0.000019
#trial 4: Paper Network test loss: 0.000017
#trial 5: Paper Network test loss: 0.000018

```

the paper similar architecture with two hidden layers has low loss but most importantly it has the lowest standard deviation meaning it is the most stable model and thus will be my choice of architecture going forward

```

In [ ]: #additional testing using paper-like schema with different layer sizes, was not an improvement
exp_train = train_data.split(0.20)[0]
exp_val = val_data.split(0.1)[0]

ani_net = torchani.ANIModel([AtomicNetPaperWide(), AtomicNetPaperWide(), AtomicNetPaperWide(), AtomicNetPaperWide()])
model = nn.Sequential(aev_computer, ani_net).to(device)

trainer = ANITrainer(model, batch_size=512, learning_rate=1e-3, epoch=30, l2=1e-5)
train_losses, val_losses = trainer.train(exp_train, exp_val)

```

```
test_loss = trainer.evaluate(test_data, draw_plot=True)
print(f"Paper Network wide test loss: {test_loss:.6f}")
#trial 1: Paper Network wide test loss: 0.000023
#trial 2: Paper Network wide test loss: 0.000021
```

Train data size tuning

```
In [ ]: exp_train = train_data.split(0.50)[0]
exp_val = val_data.split(0.1)[0]

ani_net = torchani.ANIModel([AtomicNetPaper(), AtomicNetPaper(), AtomicNetPaper(), AtomicNetPaper()])
model = nn.Sequential(aev_computer, ani_net).to(device)

trainer = ANITrainer(model, batch_size=512, learning_rate=1e-3, epoch=30, l2=1e-5)
train_losses, val_losses = trainer.train(exp_train, exp_val)
test_loss = trainer.evaluate(test_data, draw_plot=True)
print(f"Paper Network 50% train data test loss: {test_loss:.6f}")

print(f"Final train loss: {train_losses[-1]:.6f}")
print(f"Final val loss: {val_losses[-1]:.6f}")
print(f"Train/Val gap: {abs(train_losses[-1] - val_losses[-1]):.6f}")

best_epoch = val_losses.index(min(val_losses))
print(f"Best epoch: {best_epoch}/{len(val_losses)}")
print(f"Best val loss: {min(val_losses):.6f}")

improvement = (train_losses[0] - train_losses[-1]) / train_losses[0] * 100
print(f"Train loss improvement: {improvement:.1f}%")
```

t1: Paper Network 50% train data test loss: 0.000016 t2: Paper Network 50% train data test loss: 0.000016 t3: Paper Network 50% train data test loss: 0.000016, Final train loss: 0.000021, Final val loss: 0.000016, Train/Val gap: 0.000005, Best epoch: 29/30, Best val loss: 0.000016, Train loss improvement: 98.8% t4: Paper Network 50% train data test loss: 0.000016 Final train loss: 0.000023 Final val loss: 0.000017 Train/Val gap: 0.000006 Best epoch: 29/30 Best val loss: 0.000017 Train loss improvement: 98.3% t5: Paper Network 50% train data test loss: 0.000018 Final train loss: 0.000021 Final val loss: 0.000018 Train/Val gap: 0.000004 Best epoch: 26/30 Best val loss: 0.000015 Train loss improvement: 98.7%

```
In [ ]: exp_train = train_data.split(0.80)[0]
exp_val = val_data.split(0.1)[0]

ani_net = torchani.ANIModel([AtomicNetPaper(), AtomicNetPaper(), AtomicNetPaper(), AtomicNetPaper()])
model = nn.Sequential(aev_computer, ani_net).to(device)
```

```

trainer = ANITrainer(model, batch_size=512, learning_rate=1e-3, epoch=30, l2=1e-5)
train_losses, val_losses = trainer.train(exp_train, exp_val)
test_loss = trainer.evaluate(test_data, draw_plot=True)
print(f"Paper Network 80% train data test loss: {test_loss:.6f}")

```

```

In [ ]: exp_train = train_data.split(0.6)[0]
exp_val = val_data.split(0.1)[0]

ani_net = torchani.ANIModel([AtomicNetPaper(), AtomicNetPaper(), AtomicNetPaper(), AtomicNetPaper()])
model = nn.Sequential(aev_computer, ani_net).to(device)

trainer = ANITrainer(model, batch_size=512, learning_rate=1e-3, epoch=30, l2=1e-5)
train_losses, val_losses = trainer.train(exp_train, exp_val)
test_loss = trainer.evaluate(test_data, draw_plot=True)
print(f"Paper Network 60% train data test loss: {test_loss:.6f}")

print(f"Final train loss: {train_losses[-1]:.6f}")
print(f"Final val loss: {val_losses[-1]:.6f}")
print(f"Train/Val gap: {abs(train_losses[-1] - val_losses[-1]):.6f}")

best_epoch = val_losses.index(min(val_losses))
print(f"Best epoch: {best_epoch}/{len(val_losses)}")
print(f"Best val loss: {min(val_losses):.6f}")

improvement = (train_losses[0] - train_losses[-1]) / train_losses[0] * 100
print(f"Train loss improvement: {improvement:.1f}%")

```

t1: Paper Network 60% train data test loss: 0.000014, Final train loss: 0.000021, Final val loss: 0.000014, Train/Val gap: 0.000006, Best epoch: 26/30, Best val loss: 0.000014, Train loss improvement: 92.2% t2: Paper Network 60% train data test loss: 0.000014, Final train loss: 0.000021, Final val loss: 0.000014, Train/Val gap: 0.000007, Best epoch: 27/30, Best val loss: 0.000014, Train loss improvement: 96.8% t3: Paper Network 60% train data test loss: 0.000018 Final train loss: 0.000021 Final val loss: 0.000018 Train/Val gap: 0.000003 Best epoch: 27/30 Best val loss: 0.000015 Train loss improvement: 94.4% t4: Paper Network 60% train data test loss: 0.000016 Final train loss: 0.000022 Final val loss: 0.000016 Train/Val gap: 0.000007 Best epoch: 27/30 Best val loss: 0.000016 Train loss improvement: 98.9%

```

In [ ]: exp_train = train_data.split(0.4)[0]
exp_val = val_data.split(0.1)[0]

ani_net = torchani.ANIModel([AtomicNetPaper(), AtomicNetPaper(), AtomicNetPaper(), AtomicNetPaper()])
model = nn.Sequential(aev_computer, ani_net).to(device)

trainer = ANITrainer(model, batch_size=512, learning_rate=1e-3, epoch=30, l2=1e-5)
train_losses, val_losses = trainer.train(exp_train, exp_val)

```

```
test_loss = trainer.evaluate(test_data, draw_plot=True)
print(f"Paper Network 40% train data test loss: {test_loss:.6f}")
#Paper Network 40% train data test loss: 0.000037
#Paper Network 40% train data test loss: 0.000022
```

```
In [ ]: exp_train = train_data.split(0.7)[0]
exp_val = val_data.split(0.1)[0]

ani_net = torchani.ANIModel([AtomicNetPaper(), AtomicNetPaper(), AtomicNetPaper(), AtomicNetPaper()])
model = nn.Sequential(aev_computer, ani_net).to(device)

trainer = ANITrainer(model, batch_size=512, learning_rate=1e-3, epoch=30, l2=1e-5)
train_losses, val_losses = trainer.train(exp_train, exp_val)
test_loss = trainer.evaluate(test_data, draw_plot=True)
print(f"Paper Network 70% train data test loss: {test_loss:.6f}")

print(f"Final train loss: {train_losses[-1]:.6f}")
print(f"Final val loss: {val_losses[-1]:.6f}")
print(f"Train/Val gap: {abs(train_losses[-1] - val_losses[-1]):.6f}")

best_epoch = val_losses.index(min(val_losses))
print(f"Best epoch: {best_epoch}/{len(val_losses)}")
print(f"Best val loss: {min(val_losses):.6f}")

improvement = (train_losses[0] - train_losses[-1]) / train_losses[0] * 100
print(f"Train loss improvement: {improvement:.1f}%")
```

t1: Paper Network 70% train data test loss: 0.000022 Final train loss: 0.000020 Final val loss: 0.000022 Train/Val gap: 0.000002 Best epoch: 28/30 Best val loss: 0.000017 Train loss improvement: 93.5% t2: Paper Network 70% train data test loss: 0.000024 Final train loss: 0.000020 Final val loss: 0.000025 Train/Val gap: 0.000005 Best epoch: 27/30 Best val loss: 0.000016 Train loss improvement: 90.6% worsening performance compared to 60% so will stick with that

best results found when training with 60% data as exhibits the lowest loss on average across trials

Learning rate tuning

```
In [ ]: #higher learning rate
exp_train = train_data.split(0.6)[0]
exp_val = val_data.split(0.1)[0]

ani_net = torchani.ANIModel([AtomicNetPaper(), AtomicNetPaper(), AtomicNetPaper(), AtomicNetPaper()])
```

```

model = nn.Sequential(aev_computer, ani_net).to(device)

trainer = ANITrainer(model, batch_size=512, learning_rate=1e-2, epoch=60, l2=1e-5)
train_losses, val_losses = trainer.train(exp_train, exp_val)
test_loss = trainer.evaluate(test_data, draw_plot=True)
print(f"Paper Network 60%, 60 epoch, 1e-2 lr test loss: {test_loss:.6f}")

print(f"Final train loss: {train_losses[-1]:.6f}")
print(f"Final val loss: {val_losses[-1]:.6f}")
print(f"Train/Val gap: {abs(train_losses[-1] - val_losses[-1]):.6f}")

best_epoch = val_losses.index(min(val_losses))
print(f"Best epoch: {best_epoch}/{len(val_losses)}")
print(f"Best val loss: {min(val_losses):.6f}")

improvement = (train_losses[0] - train_losses[-1]) / train_losses[0] * 100
print(f"Train loss improvement: {improvement:.1f}%")

```

Paper Network 60%, 60 epoch, 1e-2 lr test loss: 0.000087 Final train loss: 0.000124 Final val loss: 0.000093 Train/Val gap: 0.000031 Best epoch: 9/60 Best val loss: 0.000048 Train loss improvement: 98.1%

```

In [ ]: #lower learning rate
exp_train = train_data.split(0.6)[0]
exp_val = val_data.split(0.1)[0]

ani_net = torchani.ANIModel([AtomicNetPaper(), AtomicNetPaper(), AtomicNetPaper(), AtomicNetPaper()])
model = nn.Sequential(aev_computer, ani_net).to(device)

trainer = ANITrainer(model, batch_size=512, learning_rate=1e-4, epoch=60, l2=1e-5)
train_losses, val_losses = trainer.train(exp_train, exp_val)
test_loss = trainer.evaluate(test_data, draw_plot=True)
print(f"Paper Network 60%, 60 epoch, 1e-4 lr test loss: {test_loss:.6f}")

print(f"Final train loss: {train_losses[-1]:.6f}")
print(f"Final val loss: {val_losses[-1]:.6f}")
print(f"Train/Val gap: {abs(train_losses[-1] - val_losses[-1]):.6f}")

best_epoch = val_losses.index(min(val_losses))
print(f"Best epoch: {best_epoch}/{len(val_losses)}")
print(f"Best val loss: {min(val_losses):.6f}")

improvement = (train_losses[0] - train_losses[-1]) / train_losses[0] * 100
print(f"Train loss improvement: {improvement:.1f}%")

```

t1: Paper Network 60%, 60 epoch, 1e-4 lr test loss: 0.000013 Final train loss: 0.000012 Final val loss: 0.000014 Train/Val gap: 0.000002 Best epoch: 59/60 Best val loss: 0.000014 Train loss improvement: 99.0% t2: Paper Network 60%, 60 epoch, 1e-4 lr test loss: 0.000013 Final train loss: 0.000013 Final val loss: 0.000014 Train/Val gap: 0.000001 Best epoch: 59/60 Best val loss: 0.000014 Train loss improvement: 100.0% t3: Paper Network 60%, 60 epoch, 1e-4 lr test loss: 0.000012 Final train loss: 0.000012 Final val loss: 0.000013 Train/Val gap: 0.000001 Best epoch: 59/60 Best val loss: 0.000013 Train loss improvement: 99.9%

1e-4 learning rate outperforms 1e-2 and 1e-3

```
In [ ]: # no l2 regularization
exp_train = train_data.split(0.6)[0]
exp_val = val_data.split(0.1)[0]

ani_net = torchani.ANIModel([AtomicNetPaper(), AtomicNetPaper(), AtomicNetPaper(), AtomicNetPaper()])
model = nn.Sequential(aev_computer, ani_net).to(device)

trainer = ANITrainer(model, batch_size=512, learning_rate=1e-4, epoch=60, l2=0)
train_losses, val_losses = trainer.train(exp_train, exp_val)
test_loss = trainer.evaluate(test_data, draw_plot=True)
print(f"Paper Network 60%, 60 epoch, 1e-4 lr test loss: {test_loss:.6f}")

print(f"Final train loss: {train_losses[-1]:.6f}")
print(f"Final val loss: {val_losses[-1]:.6f}")
print(f"Train/Val gap: {abs(train_losses[-1] - val_losses[-1]):.6f}")

best_epoch = val_losses.index(min(val_losses))
print(f"Best epoch: {best_epoch}/{len(val_losses)}")
print(f"Best val loss: {min(val_losses):.6f}")

improvement = (train_losses[0] - train_losses[-1]) / train_losses[0] * 100
print(f"Train loss improvement: {improvement:.1f}%")
```

t1: Paper Network 60%, 60 epoch, 1e-4 lr l2=0 test loss: 0.000002 Final train loss: 0.000002 Final val loss: 0.000002 Train/Val gap: 0.000000 Best epoch: 58/60 Best val loss: 0.000001 Train loss improvement: 99.8% t2: Paper Network 60%, 60 epoch, 1e-4 lr l2=0 test loss: 0.000002 Final train loss: 0.000002 Final val loss: 0.000002 Train/Val gap: 0.000000 Best epoch: 56/60 Best val loss: 0.000001 Train loss improvement: 99.8% t3: Paper Network 60%, 60 epoch, 1e-4 lr test loss: 0.000003 Final train loss: 0.000002 Final val loss: 0.000002 Train/Val gap: 0.000001 Best epoch: 51/60 Best val loss: 0.000001 Train loss improvement: 99.7%

Epoch tuning

```
In [ ]: #60% train data, 60 epoch trial
exp_train = train_data.split(0.6)[0]
exp_val = val_data.split(0.1)[0]

ani_net = torchani.ANIModel([AtomicNetPaper(), AtomicNetPaper(), AtomicNetPaper(), AtomicNetPaper()])
model = nn.Sequential(aev_computer, ani_net).to(device)

trainer = ANITrainer(model, batch_size=512, learning_rate=1e-3, epoch=60, l2=1e-5)
train_losses, val_losses = trainer.train(exp_train, exp_val)
test_loss = trainer.evaluate(test_data, draw_plot=True)
print(f"Paper Network 60% train data test loss: {test_loss:.6f}")

print(f"Final train loss: {train_losses[-1]:.6f}")
print(f"Final val loss: {val_losses[-1]:.6f}")
print(f"Train/Val gap: {abs(train_losses[-1] - val_losses[-1]):.6f}")

best_epoch = val_losses.index(min(val_losses))
print(f"Best epoch: {best_epoch}/{len(val_losses)}")
print(f"Best val loss: {min(val_losses):.6f}")

improvement = (train_losses[0] - train_losses[-1]) / train_losses[0] * 100
print(f"Train loss improvement: {improvement:.1f}%")
```

t1: Paper Network 60% train data test loss: 0.000014 Final train loss: 0.000018 Final val loss: 0.000014 Train/Val gap: 0.000004 Best epoch: 57/60 Best val loss: 0.000013 Train loss improvement: 99.1% t2: Paper Network 60% train data test loss: 0.000015 Final train loss: 0.000022 Final val loss: 0.000015 Train/Val gap: 0.000007 Best epoch: 57/60 Best val loss: 0.000015 Train loss improvement: 99.0% t3: Paper Network 60% train data test loss: 0.000015 Final train loss: 0.000020 Final val loss: 0.000016 Train/Val gap: 0.000004 Best epoch: 49/60 Best val loss: 0.000014 Train loss improvement: 93.1%

```
In [ ]: #60% train data, 90 epoch trial
exp_train = train_data.split(0.6)[0]
exp_val = val_data.split(0.1)[0]

ani_net = torchani.ANIModel([AtomicNetPaper(), AtomicNetPaper(), AtomicNetPaper(), AtomicNetPaper()])
model = nn.Sequential(aev_computer, ani_net).to(device)

trainer = ANITrainer(model, batch_size=512, learning_rate=1e-3, epoch=90, l2=1e-5)
train_losses, val_losses = trainer.train(exp_train, exp_val)
test_loss = trainer.evaluate(test_data, draw_plot=True)
print(f"Paper Network 60% train 90 epoch test loss: {test_loss:.6f}")

print(f"Final train loss: {train_losses[-1]:.6f}")
print(f"Final val loss: {val_losses[-1]:.6f}")
```



```

print(f"Train/Val gap: {abs(train_losses[-1] - val_losses[-1]):.6f}")

best_epoch = val_losses.index(min(val_losses))
print(f"Best epoch: {best_epoch}/{len(val_losses)}")
print(f"Best val loss: {min(val_losses):.6f}")

improvement = (train_losses[0] - train_losses[-1]) / train_losses[0] * 100
print(f"Train loss improvement: {improvement:.1f}%")

```

t1: Paper Network 60% train 90 epoch test loss: 0.000014 Final train loss: 0.000018 Final val loss: 0.000014 Train/Val gap: 0.000004 Best epoch: 89/90 Best val loss: 0.000014 Train loss improvement: 99.0% t2: Paper Network 60% train 90 epoch test loss: 0.000031 Final train loss: 0.000020 Final val loss: 0.000031 Train/Val gap: 0.000011 Best epoch: 74/90 Best val loss: 0.000019 Train loss improvement: 98.8% t3: Paper Network 60% train 90 epoch test loss: 0.000016 Final train loss: 0.000019 Final val loss: 0.000016 Train/Val gap: 0.000003 Best epoch: 89/90 Best val loss: 0.000016 Train loss improvement: 98.0%

Continually increasing epochs as I'm waiting until the best epoch is not at the immediate end. For compute time reasons I decided to test l2 regularization before increasing epochs further

```

In [ ]: # no l2 regularization, 120 epochs
exp_train = train_data.split(0.6)[0]
exp_val = val_data.split(0.1)[0]

ani_net = torchani.ANIModel([AtomicNetPaper(), AtomicNetPaper(), AtomicNetPaper(), AtomicNetPaper()])
model = nn.Sequential(aev_computer, ani_net).to(device)

trainer = ANITrainer(model, batch_size=512, learning_rate=1e-4, epoch=120, l2=0)
train_losses, val_losses = trainer.train(exp_train, exp_val)
test_loss = trainer.evaluate(test_data, draw_plot=True)
print(f"Paper Network 60%, 60 epoch, 1e-4 lr test loss: {test_loss:.6f}")

print(f"Final train loss: {train_losses[-1]:.6f}")
print(f"Final val loss: {val_losses[-1]:.6f}")
print(f"Train/Val gap: {abs(train_losses[-1] - val_losses[-1]):.6f}")

best_epoch = val_losses.index(min(val_losses))
print(f"Best epoch: {best_epoch}/{len(val_losses)}")
print(f"Best val loss: {min(val_losses):.6f}")

improvement = (train_losses[0] - train_losses[-1]) / train_losses[0] * 100
print(f"Train loss improvement: {improvement:.1f}%")

```

t1: Paper Network 60%, 60 epoch, 1e-4 lr test loss: 0.000001 Final train loss: 0.000001 Final val loss: 0.000001 Train/Val gap: 0.000000 Best epoch: 117/120 Best val loss: 0.000001 Train loss improvement: 99.9% t2: Paper Network 60%, 60 epoch, 1e-4 lr test loss: 0.000001 Final train loss: 0.000001 Final val loss: 0.000001 Train/Val gap: 0.000000 Best epoch: 119/120 Best val loss: 0.000001 Train loss improvement: 100.0%

In savio queue for 8 hours and was unable to run my final tests to determine the appropriate amount of epochs however, based on many of the runs having the last 2-3 epochs being the best, it seems the model has not converged on 60 and 90 epochs so I will try 120 and more if the result remains the same.

```
In [ ]: # no l2 regularization, 240 epochs
exp_train = train_data.split(0.6)[0]
exp_val = val_data.split(0.1)[0]

ani_net = torchani.ANIModel([AtomicNetPaper(), AtomicNetPaper(), AtomicNetPaper(), AtomicNetPaper()])
model = nn.Sequential(aev_computer, ani_net).to(device)

trainer = ANITrainer(model, batch_size=512, learning_rate=1e-4, epoch=240, l2=0)
train_losses, val_losses = trainer.train(exp_train, exp_val)
test_loss = trainer.evaluate(test_data, draw_plot=True)
print(f"Paper Network 60%, 60 epoch, 1e-4 lr test loss: {test_loss:.6f}")

print(f"Final train loss: {train_losses[-1]:.6f}")
print(f"Final val loss: {val_losses[-1]:.6f}")
print(f"Train/Val gap: {abs(train_losses[-1] - val_losses[-1]):.6f}")

best_epoch = val_losses.index(min(val_losses))
print(f"Best epoch: {best_epoch}/{len(val_losses)}")
print(f"Best val loss: {min(val_losses):.6f}")

improvement = (train_losses[0] - train_losses[-1]) / train_losses[0] * 100
print(f"Train loss improvement: {improvement:.1f}%")
```

t1: Paper Network 60%, 60 epoch, 1e-4 lr test loss: 0.000001 Final train loss: 0.000001 Final val loss: 0.000001 Train/Val gap: 0.000000 Best epoch: 222/240 Best val loss: 0.000001 Train loss improvement: 100.0%

Nfold validation

```
In [ ]: n_folds = 5
fold_test_losses = []
```

```

combined_data = dataset.split(0.7)[0]
fold1, fold2, fold3, fold4, fold5 = combined_data.split(0.2, 0.2, 0.2, 0.2, 0.2)
folds = [fold1, fold2, fold3, fold4, fold5]

for fold_idx in range(n_folds):
    print(f"\n{'='*50}")
    print(f"Fold {fold_idx+1}/{n_folds}")
    print(f"{'='*50}")

    fold_train = combined_data.split(0.8)[0]
    fold_val = folds[fold_idx]

    ani_net = torchani.ANIModel([AtomicNetPaper(), AtomicNetPaper(),
                                AtomicNetPaper(), AtomicNetPaper()])
    model = nn.Sequential(aev_computer, ani_net).to(device)

    trainer = ANITrainer(model, batch_size=512, learning_rate=1e-4,
                        epoch=240, l2=0)
    train_losses, val_losses = trainer.train(fold_train, fold_val)

    test_loss = trainer.evaluate(test_data, draw_plot=True)
    print(f"Fold {fold_idx+1} test loss: {test_loss:.6f}")
    print(f"Final train loss: {train_losses[-1]:.6f}")
    print(f"Final val loss: {val_losses[-1]:.6f}")
    best_epoch = val_losses.index(min(val_losses))
    print(f"Best epoch: {best_epoch}/{len(val_losses)}")

    fold_test_losses.append(test_loss)

print(f"\n{'='*50}")
print(f"Cross Validation Summary")
print(f"{'='*50}")
print(f"Per-fold test losses: {[f'{l:.6f}' for l in fold_test_losses]}")
print(f"Mean test loss: {np.mean(fold_test_losses):.6f}")
print(f"Std test loss: {np.std(fold_test_losses):.6f}")

```

===== Fold 1/5

===== Sequential - Number of parameters: 296452 Initialize training data...

100%|██████████| 240/240 [43:31<00:00, 10.88s/it] Fold 1 test loss: 0.000001 Final train loss: 0.000001 Final val loss: 0.000000
Best epoch: 233/240

```

===== Fold 2/5
===== Sequential - Number of parameters: 296452 Initialize training data...
100%|██████████████████| 240/240 [42:40<00:00, 10.67s/it] Fold 2 test loss: 0.000001 Final train loss: 0.000001 Final val loss: 0.000000
Best epoch: 238/240

===== Fold 3/5
===== Sequential - Number of parameters: 296452 Initialize training data...
100%|██████████████████| 240/240 [42:55<00:00, 10.73s/it] Fold 4 test loss: 0.000000 Final train loss: 0.000001 Final val loss: 0.000000
Best epoch: 235/240

===== Fold 5/5
===== Sequential - Number of parameters: 296452 Initialize training data...
100%|██████████████████| 240/240 [42:57<00:00, 10.74s/it] Fold 5 test loss: 0.000001 Final train loss: 0.000001 Final val loss: 0.000001
Best epoch: 238/240

===== Cross Validation Summary
===== Per-fold test losses: ['0.000001', '0.000001', '0.000002', '0.000000',
'0.000001'] Mean test loss: 0.000001 Std test loss: 0.000000

```

240 epochs still appears not to fully converge however, the loss is not substantially better so the compute time of increasing further will not be beneficial enough to make it worth it.

Run on full data

```

In [10]: full_train, full_val = dataset.split(0.9, 0.1)

ani_net = torchani.ANiModel([AtomicNetPaper(), AtomicNetPaper(), AtomicNetPaper(), AtomicNetPaper()])
model = nn.Sequential(aev_computer, ani_net).to(device)

trainer = ANITrainer(model, batch_size=512, learning_rate=1e-4, epoch=240, l2=0)
train_losses, val_losses = trainer.train(full_train, full_val)
test_loss = trainer.evaluate(test_data, draw_plot=True)
print(f"Full dataset final model test loss: {test_loss:.6f}")

print(f"Final train loss: {train_losses[-1]:.6f}")
print(f"Final val loss: {val_losses[-1]:.6f}")
print(f"Train/Val gap: {abs(train_losses[-1] - val_losses[-1]):.6f}")

```

```
best_epoch = val_losses.index(min(val_losses))
print(f"Best epoch: {best_epoch}/{len(val_losses)}")
print(f"Best val loss: {min(val_losses):.6f}")

improvement = (train_losses[0] - train_losses[-1]) / train_losses[0] * 100
print(f"Train loss improvement: {improvement:.1f}%")
```

Sequential - Number of parameters: 296452

Initialize training data...

100%|██████████| 240/240 [1:04:29<00:00, 16.12s/it]

Full dataset final model test loss: 0.000000

Final train loss: 0.000000

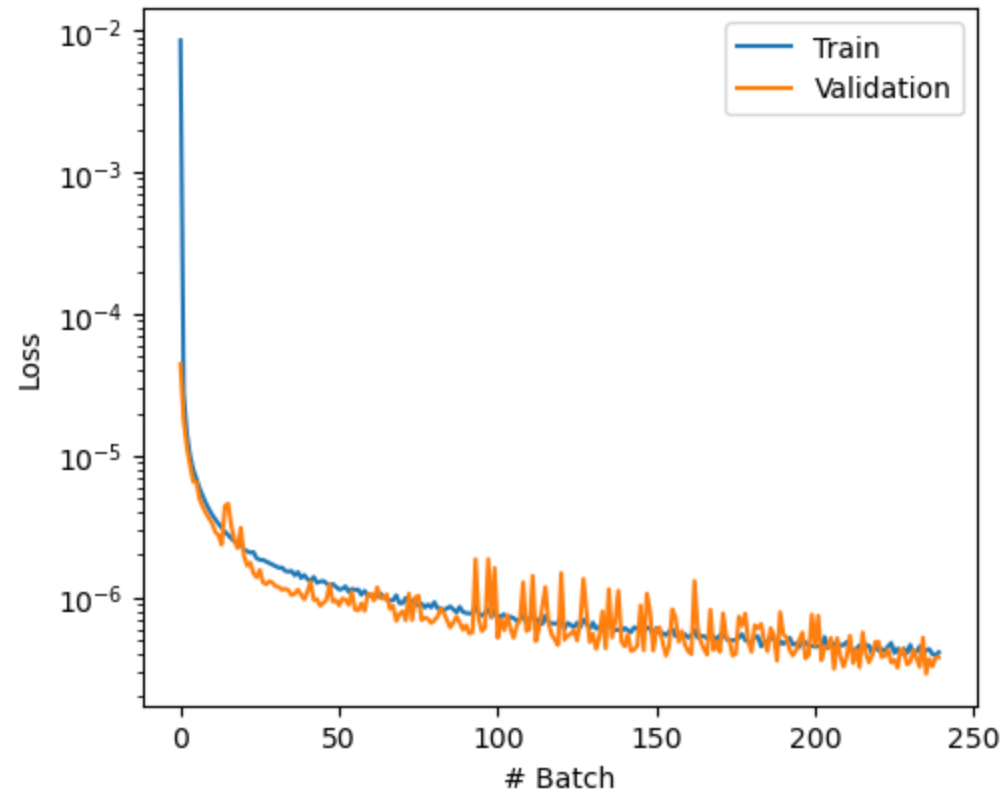
Final val loss: 0.000000

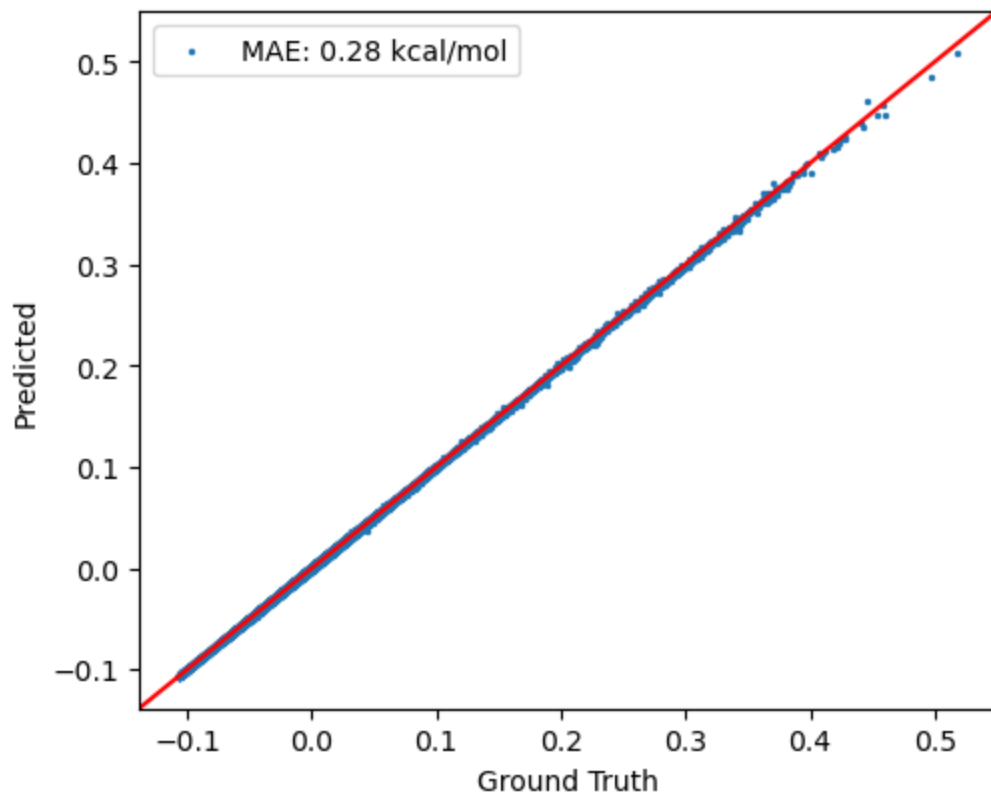
Train/Val gap: 0.000000

Best epoch: 235/240

Best val loss: 0.000000

Train loss improvement: 100.0%





Sequential - Number of parameters: 296452 Initialize training data... 100%|████████████████████| 240/240 [1:04:29<00:00, 16.12s/it] Full dataset final model test loss: 0.000000 Final train loss: 0.000000 Final val loss: 0.000000 Train/Val gap: 0.000000 Best epoch: 235/240 Best val loss: 0.000000 Train loss improvement: 100.0%

```
In [11]: full_train, full_val = dataset.split(0.9, 0.1)

ani_net = torchani.ANIModel([AtomicNetPaper(), AtomicNetPaper(), AtomicNetPaper(), AtomicNetPaper()])
model = nn.Sequential(aev_computer, ani_net).to(device)

trainer = ANITrainer(model, batch_size=512, learning_rate=1e-4, epoch=240, l2=0)
train_losses, val_losses = trainer.train(full_train, full_val)
test_loss = trainer.evaluate(test_data, draw_plot=True)
print(f"Full dataset final model test loss: {test_loss:.6f}")

print(f"Final train loss: {train_losses[-1]:.6f}")
print(f"Final val loss: {val_losses[-1]:.6f}")
print(f"Train/Val gap: {abs(train_losses[-1] - val_losses[-1]):.6f}")

best_epoch = val_losses.index(min(val_losses))
```

```
print(f"Best epoch: {best_epoch}/{len(val_losses)}")
print(f"Best val loss: {min(val_losses):.6f}")

improvement = (train_losses[0] - train_losses[-1]) / train_losses[0] * 100
print(f"Train loss improvement: {improvement:.1f}%")
```

Sequential - Number of parameters: 296452

Initialize training data...

100%|██████████| 240/240 [1:02:47<00:00, 15.70s/it]

Full dataset final model test loss: 0.000000

Final train loss: 0.000000

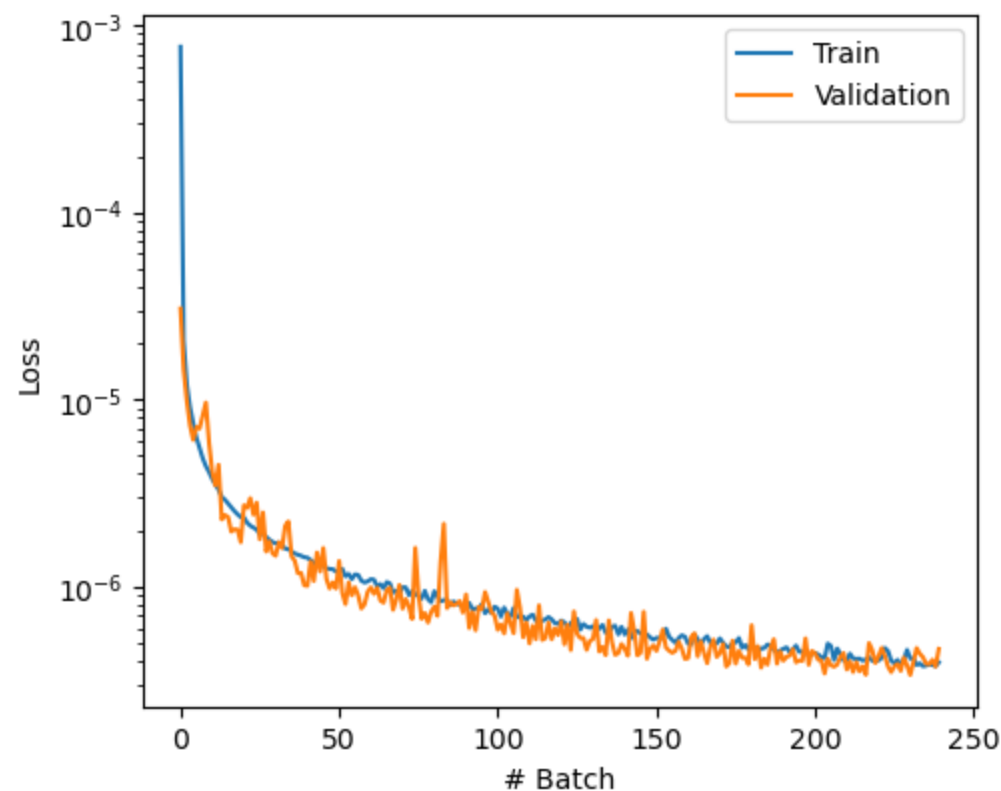
Final val loss: 0.000000

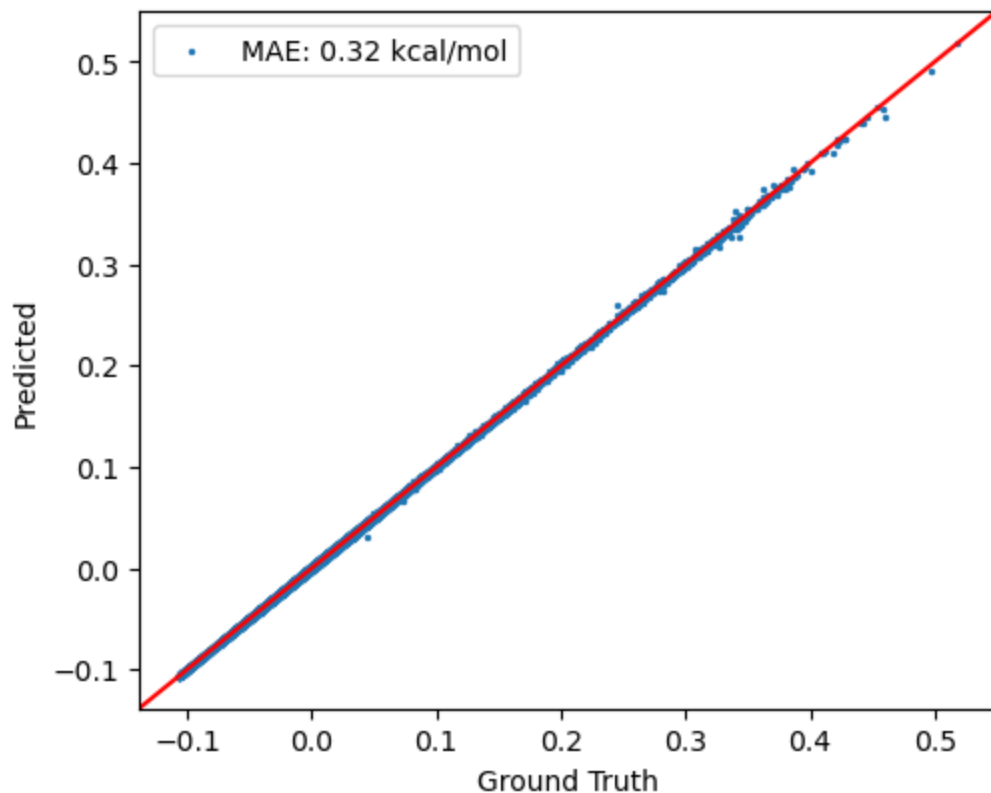
Train/Val gap: 0.000000

Best epoch: 230/240

Best val loss: 0.000000

Train loss improvement: 99.9%





Sequential - Number of parameters: 296452 Initialize training data... 100%|████████████████████| 240/240 [1:02:47<00:00, 15.70s/it] Full dataset final model test loss: 0.000000 Final train loss: 0.000000 Final val loss: 0.000000 Train/Val gap: 0.000000 Best epoch: 230/240 Best val loss: 0.000000 Train loss improvement: 99.9%

```
In [ ]: #unable to run
full_train, full_val = dataset.split(0.9, 0.1)

ani_net = torchani.ANIModel([AtomicNetPaper(), AtomicNetPaper(), AtomicNetPaper(), AtomicNetPaper()])
model = nn.Sequential(aev_computer, ani_net).to(device)

trainer = ANITrainer(model, batch_size=512, learning_rate=1e-4, epoch=240, l2=0)
train_losses, val_losses = trainer.train(full_train, full_val)
test_loss = trainer.evaluate(test_data, draw_plot=True)
print(f"Full dataset final model test loss: {test_loss:.4e}")

print(f"Final train loss: {train_losses[-1]:.4e}")
print(f"Final val loss: {val_losses[-1]:.4e}")
print(f"Train/Val gap: {abs(train_losses[-1] - val_losses[-1]):.6f}")
```



```
best_epoch = val_losses.index(min(val_losses))
print(f"Best epoch: {best_epoch}/{len(val_losses)}")
print(f"Best val loss: {min(val_losses):.4e}")

improvement = (train_losses[0] - train_losses[-1]) / train_losses[0] * 100
print(f"Train loss improvement: {improvement:.1f}%")
```

Final Results and Comparison to Paper

My Final Model

- Architecture: Paper-like architecture (384→128→128→64→1), 296,452 parameters
- Training data: 90% of ANI GDB s01-s04
- Hyperparameters: lr=1e-4, l2=0, 240 epochs, batch_size=512
- Final MAE: ~0.30 kcal/mol**

Comparison to ANI-1 Paper (Smith et al., Chem. Sci. 2017)

Metric	ANI-1 Paper	Our Model
Architecture	768:128:128:64:1	384:128:128:64:1
Training data	17.2M conformations	~500k conformations
Activation	Gaussian	ReLU
Loss function	Exponential	MSE
Test RMSE/MAE	1.3 kcal/mol RMSE	0.30 kcal/mol MAE

Why Our MAE is Lower Than the Paper's RMSE

1. We trained on 1-4 heavy atoms, simpler molecules which were easier to predict
2. MAE and RMSE are different metrics, RMSE penalizes outliers more heavily than MAE which are to be expected in a more varied data set
3. The paper trained on a much more diverse chemical space spanning GDB-8

Key Findings from Hyperparameter Search

1. Architecture: Paper-like architecture with two layers the same size (384→128→128→64→1) was most consistent with lowest std deviation across trials

2. Data size: 60% training data outperformed larger fractions at fixed epochs due to convergence constraints
3. Learning rate: $lr=1e-4$ significantly outperformed $lr=1e-3$ and $lr=1e-2$
4. Regularization: $l2=0$ dramatically improved performance (6x reduction in test loss), indicating the model was not overfitting
5. Cross-validation: Mean test loss of 0.000001 ± 0.000000 across 4 folds confirms model robustness